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# Set 7

Time: 25 minutes | Paper Reference **PMAT2/N07**

# Mathematics

## Level 2
## Section A (Non-Calculator)



**You must have:**
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks 16 /16

**My signature confirms that I will not discuss the content of the test with anyone.**

**Signature:** Master Functional Skills (MFS)

## Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of $\pi$ to be 3.14

## Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

## Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

*Turn over* ▶

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## SECTION A

**Answer ALL questions. Write your answers in the spaces provided.**

**1** Here is some data.

146 97 109 97 141 146 103 97 94 94

(a) Find the mode of this data.

**(1)**

*Frequency count: 94 (2), 97 (3), 103 (1), 109 (1), 141 (1), 146 (2) [B1]*

**• 97 has the highest frequency (3 times) --> Mode = 97**

**97** ✓

(b) Work out 17.456 – 6.072
Show your working.

**(2)**

***Pedagogical Method: Column Subtraction with Borrowing***

*Align decimal points vertically. 5 – 7 requires borrowing 1 tenth (10 hundredths).*

*borrow:* 3 15

```
  17.456
-  6.072
--------
  11.384
```

*• Thousandths: 6 – 2 = 4*
*• Hundredths: borrow 1 tenth --> 15 – 7 = 8*
*• Tenths: 3 – 0 = 3*
*• Units & Tens: 17 – 6 = 11*

*[M1] Subtraction with borrow*
*[A1] Accurate result: 11.384*

**11.384** ✓

**(Total for Question 1 is 3 marks)**

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**2** Work out $2\frac{5}{6} + 9\frac{1}{3}$

Give your answer as a mixed number.
You **must** show your working.

**(3)**

**Method 1: Improper Fractions (Direct Method)**

- 2 5/6 = (2 × 6 + 5) / 6 = 17/6
- 9 1/3 = (9 × 3 + 1) / 3 = 28/3

Step 2: Common denominator of 6 (LCM of 6 and 3 is 6):

28/3 = (28 × 2) / (3 × 2) = 56/6 [M1 common denominator]

Step 3: Add numerators over common denominator 6:

17/6 + 56/6 = (17 + 56) / 6 = 73/6 [M1 addition process]

Step 4: Convert 73/6 back to a mixed number:

73 ÷ 6 = 12 with remainder 1 --> 12 1/6 [A1 mixed number]

**Tutor Shortcut (Whole Number & Fraction Separation):**

- Whole numbers: 2 + 9 = 11
- Fractions: 5/6 + 1/3 = 5/6 + 2/6 = 7/6 = 1 1/6
- Combined total: 11 + 1 1/6 = 12 1/6 ✗

$12\frac{1}{6}$ ✓

**(Total for Question 2 is 3 marks)**

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**3** Lara wants to join a cycling club.
The club website states that the average speed of club rides is 15 miles per hour.

Lara goes on a ride to work out her average speed.
She rides 60 km in 3 hours.

1 mile = 1.6 km

Lara thinks that her average speed is lower than the average speed of club rides.

(a) Is she correct?
Show why you think this.

**(3)**

**Step 1: Calculate Lara's average speed in km/h**

**Speed = Distance ÷ Time = 60 km ÷ 3 hours = 20 km/h** *[M1]*

**Step 2: Convert Club Ride Speed (15 mph) into km/h**

**Club speed = 15 × 1.6 = 24 km/h** *[M1]*

*(15 × 1 = 15; 15 × 0.6 = 9; 15 + 9 = 24)*

*Alternative Method (Convert Lara's speed into mph):*

*Lara's speed = 20 ÷ 1.6 = 12.5 mph (Club speed = 15 mph)*

**Step 3: Compare and Conclude**

- **In km/h: 20 km/h < 24 km/h (or in mph: 12.5 < 15)**

*Yes, Lara is correct because her average speed is lower than 24 km/h.* *[A1]*

**Yes (20 km/h < 24 km/h)** ✓

☑ (b) Show a check of your answer.

**(1)**

*Part (b) Check using reverse calculation:*

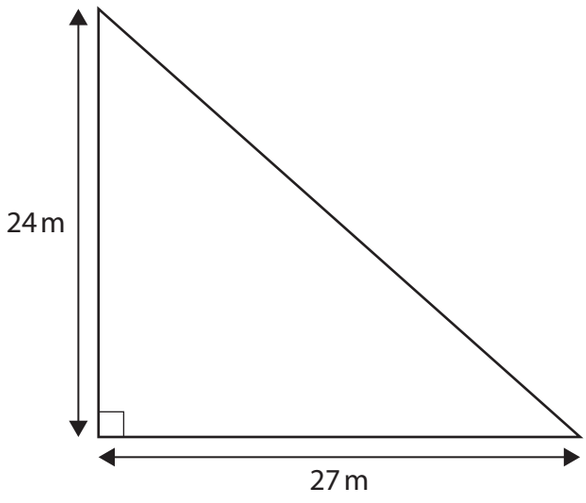
**Reverse of 15 × 1.6 = 24 km/h is 24 ÷ 1.6 = 15 mph** ☐ *[B1]*

**24 ÷ 1.6 = 15 mph** ☐ ✓

**(Total for Question 3 is 4 marks)**

**4** Bart is the manager at a building site.
He needs to lay a concrete foundation.

The diagram shows the ground space for the concrete foundation.




The depth of concrete will be 0.3 m

Concrete is delivered in concrete mixer trucks.
Each mixer truck holds 6 $m^3$ of concrete.

How many mixer trucks of concrete does Bart need in total for the concrete foundation?

**(6)**

**Step 1: Calculate the area of the triangular ground space**

Area of triangle = 1/2 × base × perpendicular height

**Area = 1/2 × 27 × 24 = 27 × 12 = 324 $m^2$** [M1 M1]

Workings: 27 × 10 = 270; 27 × 2 = 54; 270 + 54 = 324 $m^2$

**Step 2: Calculate the total volume of concrete needed**

**Volume = Area × depth = 324 $m^2$ × 0.3 m = 97.2 $m^3$** [M1]

Workings: 324 × 3 = 972; then 972 ÷ 10 = 97.2 $m^3$

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**Step 3: Calculate the number of mixer trucks needed**

Each mixer truck holds 6 $m^3$ of concrete.

**Number of trucks = 97.2 ÷ 6**

Bus Stop Division (elevated carries):

$$6 \overline{)\ 9^{3}\,7.^{1}2} \quad = 16.2 \text{ mixer trucks}$$

[M1]

**Step 4: Real-World Rounding for Mixer Trucks**

- Bart requires exactly 16.2 mixer trucks of concrete.
- Ready-mix concrete trucks cannot deliver a fraction of a truck (0.2).
- 16 trucks would only provide 96 $m^3$ (short by 1.2 $m^3$).
- **To ensure the foundation is completely filled, Bart MUST round UP to 17.**

**16.2 rounded UP --> 17 mixer trucks needed.** [A1]

**Tutor Pedagogical Summary & Golden Exam Rule:**

1. Area of triangle = (base × height) ÷ 2 = (27 × 24) ÷ 2 = 324 $m^2$
2. Volume = Area × depth = 324 × 0.3 = 97.2 $m^3$
3. 97.2 ÷ 6 = 16.2 trucks --> Must round UP to 17 whole trucks!

17

**(Total for Question 4 is 6 marks)**

**TOTAL FOR SECTION A IS 16 MARKS**

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# Set 7

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C07**

# Mathematics

## Level 2
## Section B (Calculator)



**You must have:**
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
48 /48

**My signature confirms that I will not discuss the content of the test with anyone.**

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## Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
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- Answer **all** questions.
- Write your final answers in the boxes provided.
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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a $\pi$ button take the value of $\pi$ to be 3.14

## Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
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## Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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## SECTION B

**Answer ALL questions. Write your answers in the spaces provided.**

**1** Here is a formula.

$$K = \frac{3(U + 7.15)}{V}$$

Work out the value of $K$ when $U = 2.9$ and $V = 6$

**(3)**

**Formula: K = 3(U + 7.15) / V**

*Given: U = 2.9 and V = 6*

*Step 1: Substitute U = 2.9 and V = 6 into the formula:*

**K = 3(2.9 + 7.15) ÷ 6**

*Step 2: BIDMAS – Evaluate inside the brackets first:*

**2.9 + 7.15 = 10.05** *[M1 brackets]*

*Step 3: Multiply the numerator by 3:*

**3 × 10.05 = 30.15**

*Step 4: Divide by the denominator (6):*

**K = 30.15 ÷ 6 = 5.025** *[M1 A1]*

***Tutor Tip (Simplification shortcut):***

*Notice that 3/6 simplifies to 1/2. Therefore:*

*K = (10.05) ÷ 2 = 5.025*

**5.025**

**(Total for Question 1 is 3 marks)**

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**2** Shona and Erica are gymnasts.

Here are the scores Shona got in the last 7 competitions.

| | | | | | | |
|---|---|---|---|---|---|---|
| 12.9 | 14.3 | 14.1 | 13.0 | 13.2 | 13.9 | 13.1 |

Erica only took part in six of these competitions and had a

- mean score of 13.4
- median score of 13.3
- range of scores of 1.5

Shona thinks on average her scores were better than Erica's scores.
Erica does not agree.

Explain how Shona and Erica could both be correct.
You **must** show your working.

**(3)**

**Step 1: Calculate Shona's Mean Score (7 competitions)**

Sum = 12.9 + 14.3 + 14.1 + 13.0 + 13.2 + 13.9 + 13.1 = 94.5

**Shona's Mean = 94.5 ÷ 7 = 13.5** [M1 Mean]

**Step 2: Calculate Shona's Median Score**

**Scores in order: 12.9, 13.0, 13.1, 13.2, 13.9, 14.1, 14.3**

**Middle value = 13.2 --> Shona's Median = 13.2** [M1 Median]

**Step 3: Compare both averages with Erica's results:**

- Mean: Shona (13.5) > Erica (13.4) --> Shona has higher mean
- Median: Erica (13.3) > Shona (13.2) --> Erica has higher median

**Step 4: Explanation of how BOTH can be correct:**

Shona is correct using mean; Erica is correct using median. [A1]

**They could both be correct because Shona's mean score of 13.5 is greater than Erica's mean (13.4), but Erica's median score of 13.3 is greater than Shona's median (13.2). Both mean and median are types of averages.** ✓

**(Total for Question 2 is 3 marks)**

**3** Archie puts an advert for his company online.
He has to pay £0.85 each time someone clicks on his advert.

At the end of week 1 Archie pays £650.25 for the total number of clicks on his advert.

Archie estimates that each week the total number of clicks on his advert will increase by 20% on the previous week.

Archie thinks that in week 3 more than 1000 people will click on his advert.

Is he correct?
Show why you think this.

**(4)**

**Step 1: Calculate the total number of clicks in Week 1**

Cost per click = £0.85; Total paid in Week 1 = £650.25

**Clicks in Week 1 = £650.25 ÷ £0.85 = 765 clicks** [M1 clicks]

**Step 2: Model 20% compound increase each week**

A 20% increase gives multiplier: 100% + 20% = 120% = 1.20

- **Week 2 clicks = 765 × 1.20 = 918 clicks** [M1 Wk 2]
- **Week 3 clicks = 918 × 1.20 = 1101.6 clicks** [M1 Wk 3]

Alternative compound formula: $765 \times (1.20)^2 = 765 \times 1.44 = 1101.6$

**Step 3: Compare with Archie's claim (more than 1000 clicks):**

- **Estimated clicks in Week 3 = 1101.6 (or 1102)**
- **1101.6 > 1000**

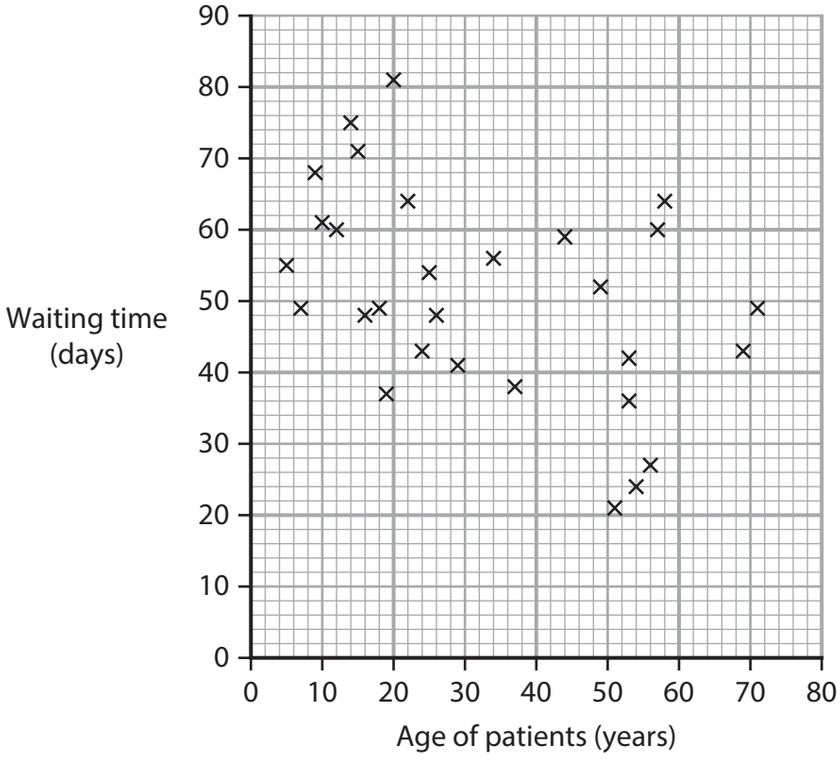
Yes, Archie is correct because 1101.6 clicks > 1000. [A1 conclusion]

**Yes (1101.6 > 1000)** ✓

**(Total for Question 3 is 4 marks)**

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**4** The scatter diagram shows information about the age of patients and the waiting time for these patients to get an appointment.




(a) Describe the relationship between the age of patients and the waiting time.

**(1)**

**Negative correlation:** [A1 correlation]

**As the age of the patient increases, the waiting time tends to decrease.**

(b) Work out the fraction of patients aged over 50 who had a waiting time greater than 30 days.

**(2)**

Step 1: Total patients aged over 50 (points right of Age = 50):

**Count: 9 patients aged over 50 (x > 50)** [M1 identify 9]

Step 2: Patients in that group with waiting time > 30 days (y > 30):

**Count: 6 patients have waiting time > 30 days**

2/3

**Step 3: Fraction = 6 / 9 = 2 / 3** [A1 6/9 or 2/3]

**(Total for Question 4 is 3 marks)**

**5** Kelly works in a grocery shop.

She wants to order 120 chocolate eggs.

Kelly finds this offer.

box of 30 chocolate eggs

normal price £65

now 16% off the normal price

She thinks she can buy 120 chocolate eggs for less than £200 with this offer.

(a) Is Kelly correct?
Show why you think this.

**(4)**

**Step 1: Calculate the number of boxes of chocolate eggs needed**

**120 eggs ÷ 30 eggs per box = 4 boxes needed** *[M1 boxes]*

**Step 2: Calculate the normal price for 4 boxes**

**Normal price = 4 × £65 = £260**

**Step 3: Apply the 16% discount**

*Discount multiplier = 100% − 16% = 84% = 0.84*

**Discounted total = £260 × 0.84 = £218.40** *[M1 discount]*

**Step 4: Compare with Kelly's claim (less than £200)**

- **£218.40 > £200 (Total cost is £18.40 over £200)**

*No, Kelly is not correct. Total cost is £218.40 (more than £200).* *[A1 conclusion]*

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*Reverse calculation check of the initial calculation:*

**Reverse of 120 ÷ 30 = 4 is: 4 × 30 = 120 eggs ☐**

**Or reverse of £218.40 ÷ 0.84 = £260; £260 ÷ 4 = £65 ☐** *[B1 check]*

**4 × 30 = 120 eggs ☐** ✓

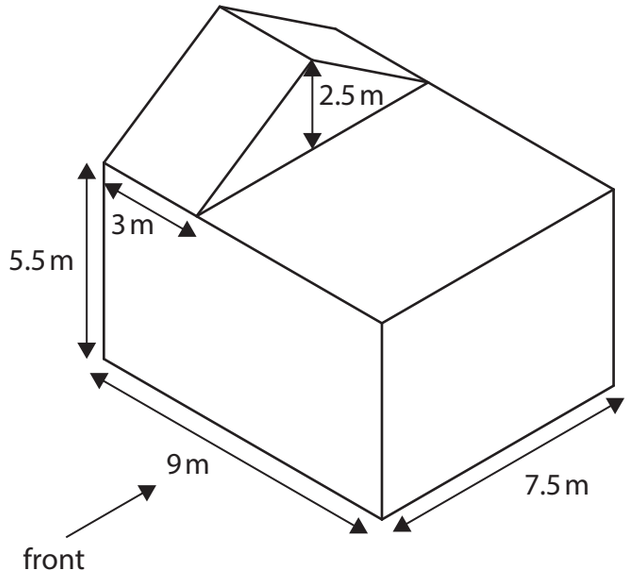
☑ (b) Use a reverse calculation to show a check of your answer.

**(1)**

**(Total for Question 5 is 5 marks)**

**6** Abdul is an apprentice architect.

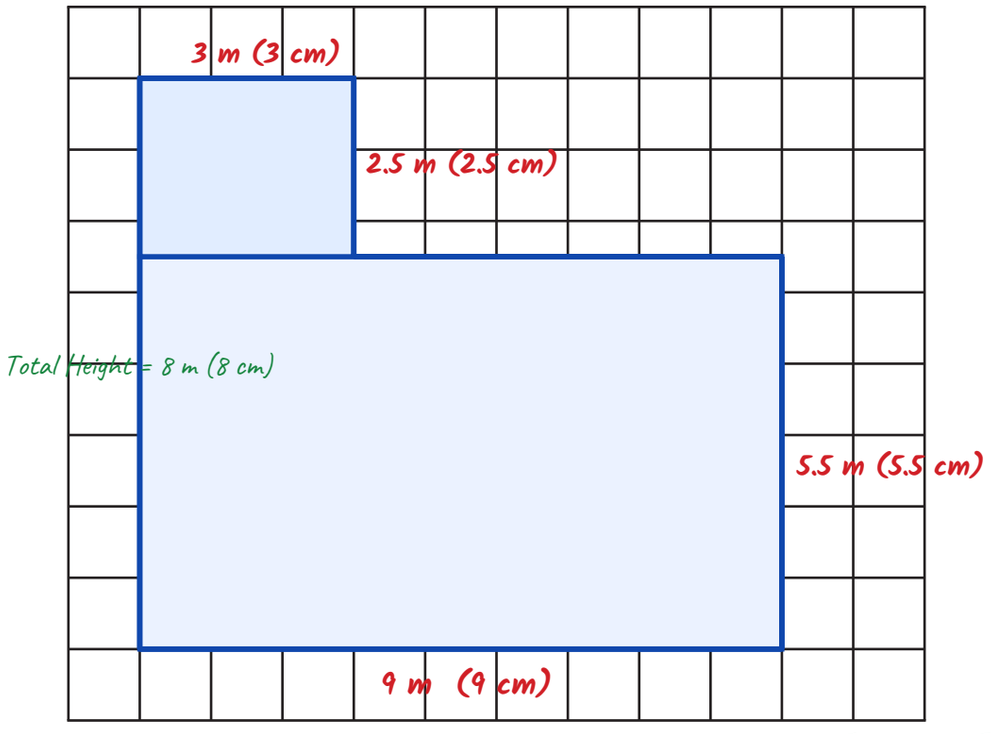
He has this diagram of a building.




Draw the front elevation of this building on the centimetre grid.
Use a scale of 1 : 100

**(3)**

Scale 1 : 100 --> 1 m in real life = 1 cm on grid (1 grid square) [M1 scale]




Full Marks: Correct front elevation to scale 1 : 100 with accurate dimensions [M1 M1 A1]

**(Total for Question 6 is 3 marks)**

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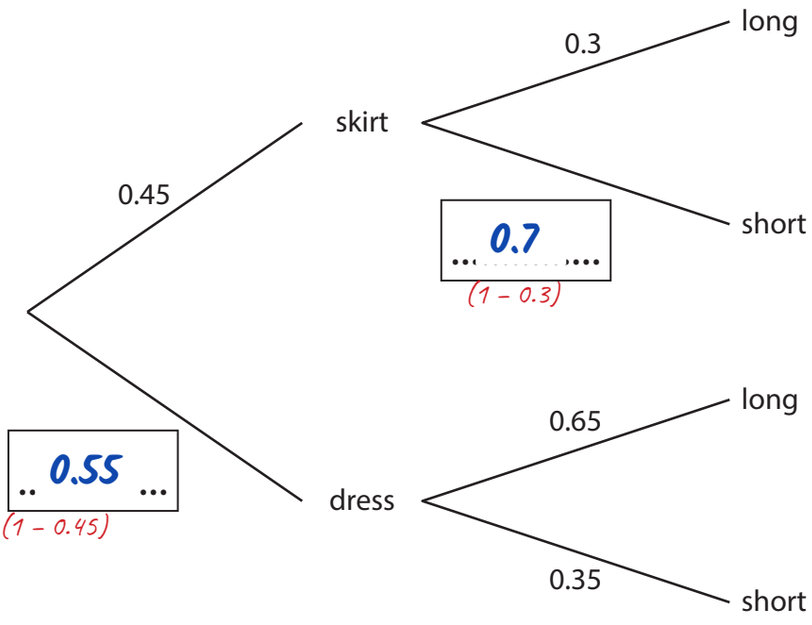
**7** The tree diagram shows the probability that an item of clothing selected at random is

- a skirt or a dress
- long or short.

(a) Complete the probability tree diagram.

[B1 B1 tree completion]

**(2)**




Golden Rule: Probabilities on each pair of branches must always sum to 1.0

• First stage: 0.45 + 0.55 = 1.0 / Skirt stage: 0.3 + 0.7 = 1.0 / Dress stage: 0.65 + 0.35 = 1.0

An item of clothing is chosen at random.

(b) Work out the probability that this item is a long skirt.

**(2)**

**Rule of Multiplication (AND Rule along the branch):**

P(Long Skirt) = P(Skirt) × P(Long | Skirt)

**P(Long Skirt) = 0.45 × 0.3 = 0.135**

[M1 process, A1 accuracy]

**0.135**

**(Total for Question 7 is 4 marks)**

**8** Takeshi is the manager at a laboratory.

The table shows information about the number of tests done at the laboratory each day in April.

| Number of tests | Frequency (days) |
|---|---|
| 1 to 10 | 8 |
| 11 to 20 | 14 |
| 21 to 30 | 5 |
| 31 to 40 | 3 |
| **Total** | **30** |

Midpoint x  f × x

5.5  8 × 5.5 = 44

15.5  14 × 15.5 = 217

25.5  5 × 25.5 = 127.5

35.5  3 × 35.5 = 106.5

Total = 495  [M1 table fx]

Takeshi expects the mean number of tests done each day in May to be 12.5% greater than the estimated mean number of tests done each day in April.

The laboratory is open for 31 days in May.

Each test brings an income of £130 for the laboratory.

Work out the expected income for the laboratory in May.
You **must** show your working.

**(6)**

**Step 1: Calculate estimated mean tests per day in April:**

Estimated Mean = $\sum fx \div \sum f = 495 \div 30 = 16.5$  [M1 Mean]

**Step 2: Calculate expected mean for May (12.5% increase):**

12.5% increase multiplier = 100% + 12.5% = 112.5% = 1.125

May Expected Mean = 16.5 × 1.125 = 18.5625 tests/day  [M1 increase]

(Or: 16.5 × 0.125 = 2.0625; 16.5 + 2.0625 = 18.5625)

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**Step 3: Total expected tests in May (31 days):** *[M1 May tests]*

Total tests in May = 18.5625 × 31 = 575.4375 tests

**Step 4: Expected total income in May (£130/test):** *[M1 total income]*

Expected Income = 575.4375 × £130 = £74,806.875

**Step 5: Format correctly as standard currency:** *[A1 accuracy]*

Rounding to 2 decimal places: £74,806.88 (or £74,807)

**Tutor Multi-Step Summary:**

1. April Estimated Mean = 495 ÷ 30 = 16.5 tests/day
2. May Daily Rate = 16.5 × 1.125 = 18.5625 tests/day
3. May Total Income = 18.5625 × 31 × £130 = £74,806.88 ✗

£ 74,806.88 ✓

**(Total for Question 8 is 6 marks)**

**9** Jasper organises large events.

Last week 23 workers took 4 hours to build a stage for a concert.

Next week Jasper wants to hire workers to build the same stage in 3 hours.

Work out the minimum number of workers needed to build the stage in 3 hours.

**(3)**

**Key Concept: Inverse Proportion**

*Fewer hours available to build means MORE workers are required to finish the job.*

**Step 1: Calculate the total work required in worker-hours:**

**Total work = 23 × 4 hours = 92 worker-hours** *[M1 work-hrs]*

**Step 2: Calculate the number of workers needed for a 3-hour build:**

**Workers needed = 92 ÷ 3 = 30.666... workers (30 2/3)** *[M1 division]*

**Step 3: Real-World Rounding Decision:**

- *Jasper cannot hire a fraction (2/3) of a worker.*
- *If he hires 30 workers: 30 × 3 = 90 worker-hours (the stage is incomplete!).*
- **To guarantee the build finishes within 3 hours, he must round UP.**

**Minimum workers needed = 31 workers** *[A1 31 workers]*

**Tutor Check & Golden Exam Rule:**

*In staffing and capacity questions, rounding down leaves work unfinished.*
*Always round up to ensure the target is met: 30.67 --> 31 workers.*

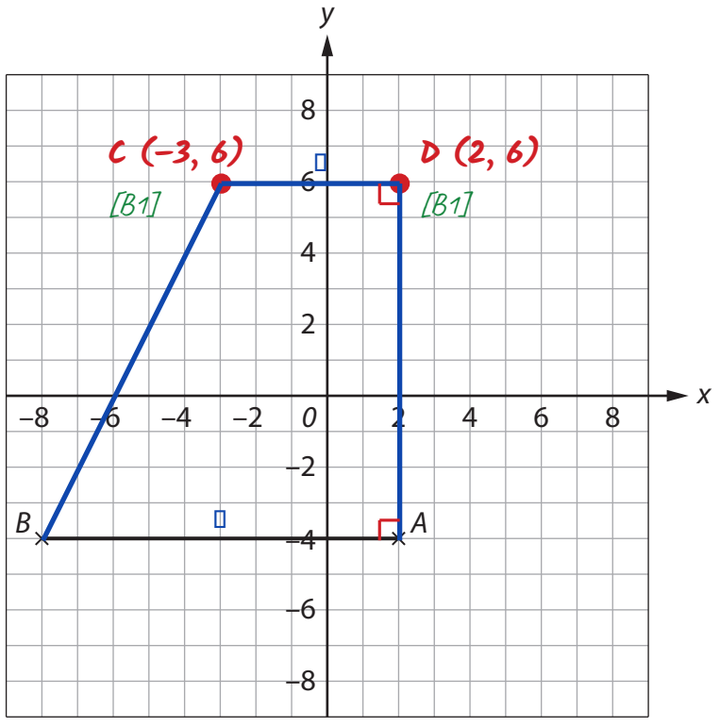
**31**

**(Total for Question 9 is 3 marks)**

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**10** Here is a coordinate grid.




(a) Plot and label the point $C$ at $(-3, 6)$

**(1)**

Sylvie wants to draw a trapezium $ABCD$ on the grid.

She will use the straight line $AB$ as the base of the trapezium.
Sylvie wants the trapezium to have two right angles.

(b) Plot and label a point $D$ on the grid to complete the trapezium for Sylvie.

**(2)**

***Pedagogical Explanation for Sylvie's Trapezium:***

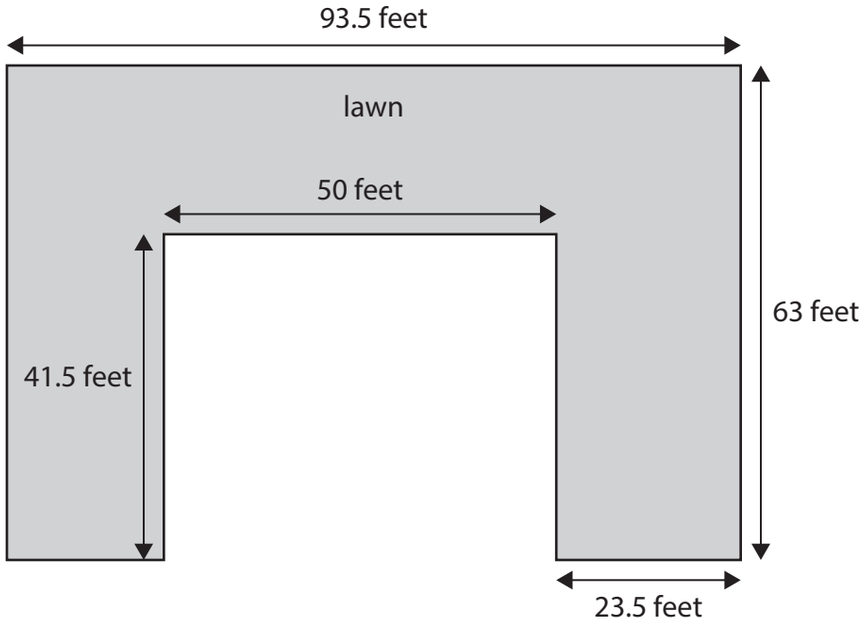
- *Base AB is on horizontal line y = −4.*
- *A vertical line from A(2, −4) up to D(2, 6) along x = 2 creates the first 90° right angle.*
- *A horizontal line from C(−3, 6) to D(2, 6) along y = 6 creates the second 90° right angle.*
- *AB is parallel to CD, so ABCD is a right-angled trapezium with two right angles. [M1 A1]* ☒

**(Total for Question 10 is 3 marks)**

**11** Vicky works in a park.

She needs to put edging around a lawn in the park.

The diagram shows the dimensions of the lawn.
The lawn is made up of rectangles.




Vicky will buy edging in 5 metre lengths.
She can cut and join the lengths of edging.
She knows that 1 metre is 3.3 feet.

Each length of edging costs £38.99
Vicky has £1000 to buy the edging.

Does Vicky have enough money to buy all the edging she needs?

**(5)**

**Step 1: Calculate the total perimeter of the lawn in feet**

Missing bottom left width = 93.5 – 50 – 23.5 = 20 feet

Perimeter = 93.5 + 63 + 23.5 + 41.5 + 50 + 41.5 + 20 + 63

**Total Perimeter = 396 feet** [M1 perimeter]

**Step 2: Convert perimeter from feet into metres (1 m = 3.3 ft)**

**Perimeter in metres = 396 ÷ 3.3 = 120 metres** [M1 conversion]

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**Step 3: Number of 5 m edging lengths needed:** *[M1 lengths]*

**Number of lengths = 120 metres ÷ 5 m = 24 lengths**

**Step 4: Calculate the total cost of the edging lengths:** *[M1 total cost]*

**Total Cost = 24 × £38.99 = £935.76**

*(Workings: 24 × 39 = 936; 936 – (24 × 0.01) = £935.76)*

**Step 5: Compare with Vicky's budget of £1000:**

- Vicky has £1000 to buy the edging.
- **£935.76 < £1000** *(Remaining change: £1000 – £935.76 = £64.24)*

*Yes, Vicky has enough money to buy all the edging she needs.* *[A1 conclusion]*

**Tutor Multi-Step Summary:**

1. Perimeter = 396 ft --> 396 ÷ 3.3 = 120 m
2. Edging lengths = 120 ÷ 5 = 24 lengths
3. Cost = 24 × £38.99 = £935.76 < £1000 --> Enough money!

**Yes (£935.76 < £1000)** ✓

**(Total for Question 11 is 5 marks)**

**12** (a) Write as a number

seven million four hundred and thirty thousand nine hundred.

**(1)**

**7 430 900** [B1]

Dorothy reads an article about a talent show.

The article states that 17 424 people applied to enter the show last year.
Each person that applied was either a singer or a dancer.

The ratio of the number of singers to the number of dancers was 3 : 5

This year 19 500 people applied to enter the show.
39% of these people were singers.

Dorothy thinks that at least 1200 more singers applied to enter the show this year than last year.

(b) Is she correct?
Show why you think this.

**(5)**

**Step 1: Calculate the number of singers last year**

Applicants last year = 17 424; Ratio singers : dancers = 3 : 5

Total ratio parts = 3 + 5 = 8 parts

**1 part = 17 424 ÷ 8 = 2178 people** [M1 ratio parts]

**Singers last year = 3 × 2178 = 6534 singers** [M1 singers]

**Step 2: Calculate the number of singers this year**

Total applicants this year = 19 500; 39% were singers

**Singers this year = 0.39 × 19 500 = 7605 singers** [M1 percentage]

(Mental check: 40% of 19500 = 7800; 1% = 195; 7800 – 195 = 7605)

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**Step 3: Calculate the difference in singers between the two years:**

Difference = Singers this year − Singers last year

**Difference = 7605 − 6534 = 1071 singers** [M1 difference]

**Step 4: Compare with Dorothy's claim (at least 1200 more singers):**

- Dorothy thinks at least 1200 more singers applied.
- **1071 < 1200 (short of Dorothy's claim by 129 singers)**

No, Dorothy is not correct. Only 1071 more singers applied this year, which is less than 1200. [A1 conclusion]

**Tutor Multi-Step Summary:**

1. Last year: 17 424 ÷ 8 × 3 = 6534 singers
2. This year: 19 500 × 0.39 = 7605 singers
3. 7605 − 6534 = 1071 < 1200 --> Dorothy is incorrect! ✗

**No (1071 < 1200)** ✓

**(Total for Question 12 is 6 marks)**

**TOTAL FOR SECTION B IS 48 MARKS**
**TOTAL FOR PAPER IS 64 MARKS**





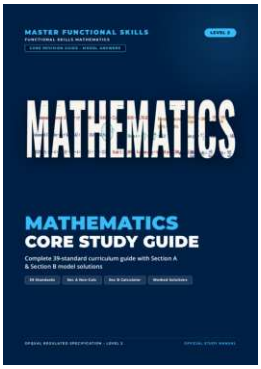
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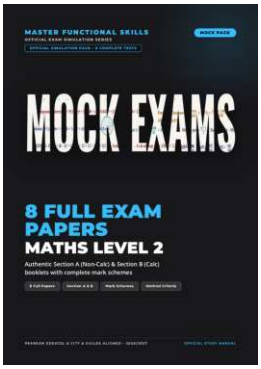



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